

SHREEYAN RIJAL

Computational Materials Physicist / Prospective PhD Researcher — Photovoltaic & Energy Materials

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Research Profile

Computational materials physicist specializing in first-principles (DFT) modeling of photovoltaic and energy-conversion materials. Work spans electronic-structure calculations, spin-polarized transport, and device-level simulation to study stability, efficiency, and optoelectronic behavior in perovskite solar cells, doped semiconductors, and van der Waals heterostructures. Author of four first-author peer-reviewed publications in condensed matter and photovoltaic materials physics, with additional first-author and co-authored work in progress across materials chemistry and computational biophysics, moving independently from first-principles theory through simulation to publishable results. Seeking a fully funded PhD position to develop computational and data-driven methods for next-generation solar cell and quantum/energy materials.

Education

B.Sc. Physics

2021–2025

St. Xavier's College, Tribhuvan University • Kathmandu, Nepal

Relevant Coursework: Quantum Mechanics, Condensed Matter Physics, Electrodynamics, Thermodynamics, Computational Methods

Research Experience

- Performed DFT modeling of perovskite and thin-film solar cell architectures in Quantum ESPRESSO under simulated thermal and optical stress.
- Ran complementary device-level simulations in OghmaNano to identify parameter regimes that improve stability and efficiency.
- Carried out spin-polarized first-principles calculations of Mn-doped silicon, resolving impurity-induced near-metallicity.
- Characterized enhanced infrared optical activity in the same system, with direct relevance to spintronic device design.
- Modeled strain-engineered optoelectronic and thermoelectric response of two van der Waals heterostructures: Janus MoSSe/Ti₂CO₂ and Zr₂CO₂/blue phosphorus.
- Built Python-based scientific computing workflows for data analysis and machine-learning-assisted interpretation of simulation and experimental datasets.
- Collaborated across multi-institutional research teams, translating first-principles results into four first-author peer-reviewed publications.

Publications (Peer-Reviewed, First Author)

1. **Rijal, S.** Strain-tunable band alignment, optical anisotropy, and thermoelectric response of a Janus MoSSe/Ti₂CO₂ van der Waals heterostructure. *Journal of Physics and Chemistry of Solids*, 114075. (Sole Author) DOI: [10.1016/j.jpcs.2026.114075](https://doi.org/10.1016/j.jpcs.2026.114075)
2. **Rijal, S.**, Chaudhary, P., Sharma, S., Thada Magar, M.T., Regmi, B., Adhikari, B., et al. Time-frequency characterization of prompt penetration electric fields across distinct geomagnetic regimes. *Advances in Space Research*. DOI: [10.1016/j.asr.2026.06.060](https://doi.org/10.1016/j.asr.2026.06.060)
3. **Rijal, S.**, Jha, V.K., et al. Impurity-induced near-metallicity, spin-polarized transport, and enhanced infrared optical activity in Mn-doped silicon: A spin-polarized first-principles study. *Physica B: Condensed Matter*, 739, 418967. DOI: [10.1016/j.physb.2026.418967](https://doi.org/10.1016/j.physb.2026.418967)
4. **Rijal, S.**, Sah, N., Chaudhary, P., & Jha, V.K. Enhancing the stability and efficiency of perovskite solar cells by optimization of electrical parameter under thermal and optical stress. *Optical and Quantum Electronics* (Springer), 58, 288. DOI: [10.1007/s11082-026-08876-3](https://doi.org/10.1007/s11082-026-08876-3)

Ongoing Projects

1. **Rijal, S.** Strain-Engineered Optoelectronic and Thermoelectric Properties of Zr_2CO_2 /Blue Phosphorus van der Waals Heterostructure: A First-Principles Study. (Sole/Lead Author, Under Review)
2. Rijal, S., et al. Relativistic Orbital Hybridization, Anisotropic Charge Transport and Dielectric Screening in Lead-Free Layered Double Perovskite $\text{Cs}_4\text{MnSb}_2\text{Cl}_{12}$ for Autonomous Indoor Photovoltaics. *Advanced Energy Materials*. (Co-Author, Under Review)
3. **Rijal, S.** Conceptual DFT Analysis of Luteolin: Frontier Orbitals, Fukui Functions, and Electrostatic Potential Mapping of the Antioxidant Pharmacophore. (Lead Author, In Preparation)
4. Rijal, S., et al. Directional Spin–Orbit Coupling and Nonlinear Optical Response in a Lead-Free Bismuth–Silver Perovskite Cluster. (Co-Author, In Preparation)

Technical Skills

DFT & Simulation: Quantum ESPRESSO, Gaussian, OghmaNano (device simulation), ORCA (collaborator)

Programming & Analysis: Python (scientific computing, machine learning, automation), MATLAB, C, $\text{\LaTeX}$

Physics Areas: Condensed matter and semiconductor physics, electronic structure, spintronics, photovoltaics, optoelectronics, thermoelectrics

Leadership & Mentorship

- **Research Mentor**, St. Xavier's College (2023–Present) — Guide junior undergraduates into DFT research and scientific writing.
- **LSRA Mentor** (2024–Present) — Mentor a candidate for the Fathers Locke & Stiller Research Award.
- **Library Council Representative**, Physics Department, St. Xavier's College (2021–2025) — Sustained institutional leadership and service throughout undergraduate study.

Languages

Nepali (Native) • English (Fluent, Professional Proficiency) • Hindi (Fluent, Professional Proficiency)

References

Dr. Vinaya Kumar Jha — Research Supervisor
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Dr. Binod Adhikari — Research Supervisor; Former Head of Department
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Dr. Basu Dev Ghimire — Research Supervisor; Head of Department
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